CH365 Chemical Engineering Thermodynamics

Lesson 26
Review

Homework

Problem 5.50 part 1

Ethylene vapor is cooled at atmospheric pressure from 830 to 35 °C by direct heat transfer to the surroundings at 25 °C. With respect to this surroundings temperature, what is the lost work of the process in kJ/mol?

The heat capacity of ethylene is given in Table C.1 of Appendix C.

Problem 5.50, continued

part 2

Show that the same result is obtained as the work which can be derived from a reversible heat engine operating with ethylene vapor as the heat source and the surroundings as the sink.

Problem 5.17

A Carnot engine operates between temperature levels of 600 K and 300 K. It drives a Carnot refrigerator, which provides cooling at 250 K and discards heat at 300 K.

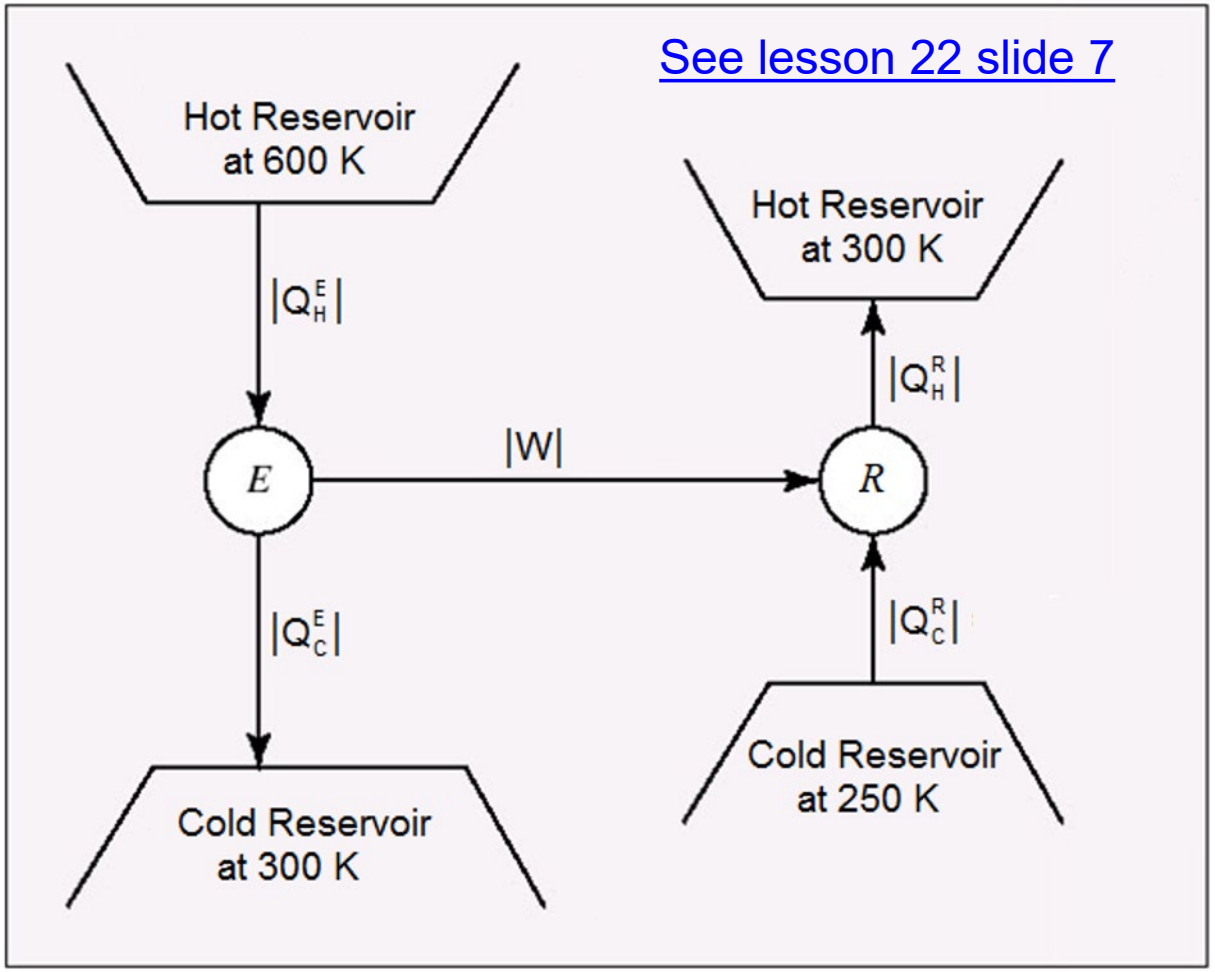
Determine a numerical value for the ratio of heat extracted by the refrigerator ("cooling load") to the heat delivered to the engine ("heating load").

"Q_C^R"

"Q_H^E"

Problem 5.17, continued

Determine a numerical value for the ratio of heat extracted by the refrigerator ("cooling load") to the heat delivered to the engine ("heating load").



Want relationship between Q_C^R and Q_H^E

Problem 5.17, continued

Carnot Engine	Carnot Heat Pump (Refrigerator)
Efficiency, η	Coefficient of Performance, COP
absorbs $\lvert Q_H^E \rvert$, discards $\lvert Q_C^E \rvert$, and produces $\lvert W \rvert$	absorbs $\lvert Q_C^R \rvert$, discards $\lvert Q_H^R \rvert$, and consumes $\lvert W \rvert$
$W + Q_H^E + Q_C^E = 0$ $<0 \quad >0 \quad <0$ $-\lvert W \rvert + \lvert Q_H^E \rvert - \lvert Q_C^E \rvert = 0$ $\lvert W \rvert = \lvert Q_H^E \rvert - \lvert Q_C^E \rvert$ 1st Law	$W + Q_H^R + Q_C^R = 0$ $>0 \quad <0 \quad >0$ $\lvert W \rvert - \lvert Q_H^R \rvert + \lvert Q_C^R \rvert = 0$ $\lvert W \rvert = \lvert Q_H^R \rvert - \lvert Q_C^R \rvert$ 1st Law
work output ÷ heat "pulled" output / input $\eta \equiv \dfrac{\lvert W \rvert}{\lvert Q_H^E \rvert}$ (Eq. 5.6) definition of efficiency $\eta < 0$ input > output	work input ÷ heat "pulled" $\eta_{refrig} \equiv \dfrac{\lvert W \rvert}{\lvert Q_C^R \rvert}$ output / input $COP \equiv \dfrac{\lvert Q_C^R \rvert}{\lvert W \rvert}$ definition of coefficient of performance (or COP) COP > 0 output > input
$\eta = \dfrac{\lvert Q_H^E \rvert - \lvert Q_C^E \rvert}{\lvert Q_H^E \rvert} = 1 - \dfrac{\lvert Q_C^E \rvert}{\lvert Q_H^E \rvert}$	$COP = \dfrac{\lvert Q_C^R \rvert}{\lvert Q_H^R \rvert - \lvert Q_C^R \rvert} \Rightarrow \dfrac{1}{COP} = \dfrac{\lvert Q_H^R \rvert - \lvert Q_C^R \rvert}{\lvert Q_C^R \rvert} = \dfrac{\lvert Q_H^R \rvert}{\lvert Q_C^R \rvert} - 1$
Carnot's equation $0 = \dfrac{Q_H^E}{T_H^E} + \dfrac{Q_C^E}{T_C^E}$ $\dfrac{\lvert Q_C^E \rvert}{\lvert Q_H^E \rvert} = \dfrac{T_C^E}{T_H^E}$ (Eqns. 5.4 L22 Slide 8)	Carnot's Equation $0 = \dfrac{Q_H^R}{T_H^R} + \dfrac{Q_C^R}{T_C^R}$ $\dfrac{\lvert Q_H^R \rvert}{\lvert Q_C^R \rvert} = \dfrac{T_H^R}{T_C^R}$
$\eta = 1 - \dfrac{T_C^E}{T_H^E}$ (Eq. 5.7)	$\dfrac{1}{COP} = \dfrac{T_H^R}{T_C^R} - 1 = \dfrac{T_H^R - T_C^R}{T_C^R} \Rightarrow COP = \dfrac{T_C^R}{T_H^R - T_C^R}$